



Assessment of driving safety and comfort during vortex-induced vibrations in a long-span bridge considering wind-vehicle-bridge interactions

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ABSTRACT

Vortex-induced vibration (VIV) has a detrimental influence on the traffic management of bridges and threatens driving safety and comfort. To quantitatively assess driving safety and comfort under VIV conditions, this study incorporates a nonlinear vortex-excited force aerodynamic model to simulate VIVs based on the classic theory of wind-vehicle-bridge interactions. Based on the wheel load reduction rate and acceleration response to vehicle vibration, driving safety and comfort are analyzed from the perspective of vehicle dynamic performance. A novel approach is firstly proposed to quantify the influence of blind area on driving safety by incorporating the concept of stopping sight distance, along with the introduction of a general fitting formula to address this issue. Additionally, the factors influencing drivers' vision under VIV are discussed, including height, distance and time-history. This study provides a comprehensive comparison of the permitted limits on VIV amplitudes from the perspectives of driving safety and comfort, highlighting the significance of considering the comfort index based on vehicle dynamic response. This approach effectively addresses the shortcomings of existing specifications in assessing driving comfort while proposing formulations for calculating VIV amplitude limits at different vehicle speeds, aiming to increase the maximal transportation capacity.

1. Introduction

As the span of modern bridges continues to increase, their damping and rigidity are decreasing, making them more sensitive to the effects of wind. With respect to vortex-induced vibration (VIV), an increasing bridge span will lead to a decrease in the VIV lock-in range of the wind speed, increasing the probability of experiencing VIVs at low wind speeds. Wu et al. (2023) and Zhao et al. (2023) have reported many studies on the mechanism of VIVs and torsional and vertical VIV phenomena in superlong span bridges. Specifically, they noted that the VIV performance could be deteriorated by nonstructural elements, such as wind barriers and maintenance rails. However, notable events of large-amplitude VIVs still occur globally, such as for the Great Belt East Bridge in Denmark (Larsen et al., 2000), the Rio-Niteroi Bridge in Brazil (Battista and Pfeil, 2000), and the Trans-Tokyo Bay Highway Crossing in Japan (Fujino and Yoshida, 2002). Recently, many cases of severe VIVs have occurred in China, such as at the Xihoumen Bridge in Zhoushan (Li et al., 2014), the Humen Bridge in Guangzhou (Ge et al.,

2022; Zhao et al., 2022) and the Yingwuzhou Bridge in Wuhan (Qin et al., 2024). Among them, traffic control measures were implemented on the Humen Bridge due to the large-amplitude VIV, attracting considerable attention from society. Because of the self-limiting load and vibration amplitude, VIVs usually cannot cause structural collapse. However, excessively large vibration amplitudes can affect driving safety and comfort and have attracted severe public concern (Larsen et al., 2000; Chen and Huang, 2015; Cao et al., 2020). VIVs typically occur at low wind speeds and have a relatively high probability of occurrence, emphasizing the necessity of assessing driving safety and comfort during VIVs. However, related research is still lacking.

The evaluation of driving safety and comfort during VIVs can be conducted from two perspectives: vehicle dynamic response (Zhang et al., 2021; Guruguntla et al., 2023; Deubel et al., 2023) and driver vision (Chen and Huang, 2015; Zhu et al., 2023). Various indices have been widely utilized to assess driving safety and driving comfort based

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on vehicle dynamic responses. Regarding the safety index based on the vehicle dynamic response, Baker (1987) initially quantified vehicle accidents and proposed three accident indices, namely, overturning, side-slip, and rotational accidents, which are widely used to evaluate driving safety (Guo and Xu, 2006; Guo and Ma, 2024). Other driving safety indices, such as the roll safety factor (RSF) (Liu, 1999) and load transfer ratio (LTR) (Wang and Xu, 2015), were developed based on Baker's indices. For indices related to driving comfort based on vehicle dynamic response, regulations differ by country. In China, the 'Wind-Resistant Design Specification for Highway Bridges' (JTG/T 3360-01-2018) stipulates that the peak vertical acceleration should not exceed 3.6 m/s^2 , which refers to the international evaluation indicator ISO 2631-1:1997 (Mechanical Vibration, 1997). It introduces the 'Overall Vibration Total Value' to assess human comfort of whole-body vibration through frequency-weighted evaluation, which is widely applied to evaluate driving comfort (Qin et al., 2024; Zhou and Chen, 2016; Zhu et al., 2021).

Regarding driving safety and comfort based on a driver's vision during VIVs, most research employs methods that utilize the simplified method proposed by Chen and Huang (2015). In 2001, Larsen et al. (2000) first proposed that undulating road surfaces might affect the driver's vision when studying VIVs on the Great Belt East Bridge. Chen and Huang (2015) presented a simplified method to calculate the blind area during VIVs. The results indicated that when the maximum height of the blind area is less than 0.1 m, the VIV amplitude should be less than 0.35 m. Zhu et al. (2023) applied this method to calculate the changes in the height and duration of blind areas during VIVs involving different types of vehicles. Zeng et al. (2022) used numerical methods to calculate the length variation of the front blind area of vehicles during VIVs. Existing researches have not clearly defined the impact of blind area on driving safety. It is commonly considered dangerous when the height of blind area exceeds 0.1 m, while the distance between the driver and the blind area is often overlooked. In fact, a large blind area can still be considered safe as long as it is located far enough from the driver. Additionally, current research lacks information on the influence of blind areas on driving comfort.

According to the wind resistance design specification (Ministry of Transports of the People's Republic of China, 2018), the VIV amplitude permitted limits are suitable only for bridges with spans of less than 200 m. The VIV amplitude is determined based on the acceleration amplitude, with the aim of establishing amplitude limits according to driving comfort. However, the method proposed by the specification directly adopts the simplified assumption that bridge deck acceleration is equal to drivers' acceleration, lacking intermediate variables such as driving speed and vehicle elastic suspension system, thereby limiting its applicability. Currently, traffic control standards for significant bridge vibrations rely heavily on empirical assessment, and a framework for theoretical analysis or wind tunnel testing is lacking. Therefore, managers must adopt relatively conservative standards for traffic control, leading to the inefficient use of existing traffic resources. However, if the standards are too rigorous, the difficulty of vehicle control on the bridge may increase, affecting the comfort and subjective psychological perception of occupants. Under extreme conditions, this consequence can easily lead to side-slip, overturning or rotational accidents. In summary, a reasonable assessment of driving safety and comfort under the influence of VIVs and a scientific determination of traffic restriction standards are urgently needed for engineering applications.

Currently, the wind-vehicle-bridge interaction (WVBI) calculation framework has been substantially refined (Cai and Chen, 2004; Han et al., 2014; Camara et al., 2019). However, existing research has focused primarily on the impact of bridge buffeting vibrations on driving safety under high crosswind speeds, and an in-depth understanding of the effects of VIVs on driving safety and comfort within the wind speed lock-in range is lacking. Furthermore, few discussions of the influence of the driver's vision during WVBI are available. This study evaluated driving safety and comfort during VIVs from the perspectives

of vehicle dynamic response and driver vision. By calculating the dynamic response of vehicle models on a large-span bridge during VIV processes, various factors, including vehicle speed, VIV mode, and VIV amplitude, are quantitatively compared. Using numerical methods, this study defines the safety index and proposes a fitting formula for the safety index based on driver vision VIVs. This approach precisely evaluates the influence of blind areas on the driving safety of occupants. Finally, reasonable equations for VIV amplitude limits that consider comprehensive factors from the perspective of both the vehicle dynamic response and the driver's vision are proposed, and the problem of excessively strict VIV amplitude limits provided by specifications at low vehicle speeds while neglecting risks at high speeds is also addressed.

2. Wind-vehicle-bridge interaction under VIVs

2.1. Wind-vehicle-bridge interaction modeling

This study established a 13-degree-of-freedom (DOF) vehicle dynamic model to calculate the dynamic response of a vehicle. The vehicle dynamic model is a combination of one rigid body connected by four axle mass blocks and a series of springs and damping devices, as shown in Fig. 1. The vehicle is a Sedan car model whose parameters were referenced from the document (Zhou and Chen, 2016) and are listed in Table 1. The DOFs are defined in Eq. (1) and include 5 rigid DOFs and 8 wheel DOFs.

$$Z = [Z_v, Y_v, \theta_v, \phi_v, \varphi_v, Z_{s1}, Y_{s1}, Z_{s2}, Y_{s2}, Z_{s3}, Y_{s3}, Z_{s4}, Y_{s4}] \quad (1)$$

In Eq. (1), Z_v represents the vehicle's vertical vibration, Y_v is the lateral vibration of the vehicle, θ_v is the pitching vibration of the rigid body, ϕ_v is the rolling vibration of the rigid body and φ_v is the yawing vibration of the rigid body. Z_{si} ($i = 1, 2, 3, 4$) represents the vertical vibration of the four wheels, and Y_{si} ($i = 1, 2, 3, 4$) represents the lateral vibration of the four wheels.

By introducing the vortex-excited force into the classic wind-vehicle-bridge coupled vibration equations, motion equations under VIV conditions are established, as shown in Eqs. (2a) and (2b).

$$M_b \ddot{u}_b + C_b \dot{u}_b + K_b u_b = F_{wb} + F_{vb} \quad (2a)$$

$$M_v \ddot{u}_v + C_v \dot{u}_v + K_v u_v = F_{wv} + F_{bv} \quad (2b)$$

In Eqs. (2a) and (2b), M , C , and K represent the mass, damping and stiffness matrices, respectively, with subscripts b and v denoting the bridge and vehicle, respectively. Among them, M_b and K_b are model mass and stiffness matrices generated by FEM in ANSYS. The damping ratios of all bridge modes are set to 3%. F_{wb} represents the aerodynamic forces on the bridge, which are simulated using the Scanlan empirical nonlinear model (Ehsan and Scanlan, 1990). During VIVs, the wind speed is generally less than 10 m/s with a low turbulence intensity. In this study, a mean wind speed of 8 m/s and turbulence intensity of 0.1 are adopted. The wind speed is generated using WAWS (Cao et al., 2000), while the aerodynamic forces on the vehicle are calculated using the quasistatic method. The aerodynamic force coefficients for the vehicle are based on the fitting results of Coleman and Baker (1990). F_{vb} and F_{bv} represent the interaction forces between the vehicle and the bridge, which are the interaction forces calculated in Eqs. (2a) and (2b). The Newmark- β method with a time step of 0.01 s is used for the computation in the time domain. The flow chart of the calculation framework is shown in Fig. 2.

2.2. Vortex-excited force

The primary objective of our calculation framework is to reproduce the bridge's vibration behavior in the form of a sine wave. During VIV, the average wind speed is typically moderate, resulting in minimal contributions from buffeting and self-excited forces to the overall vibration

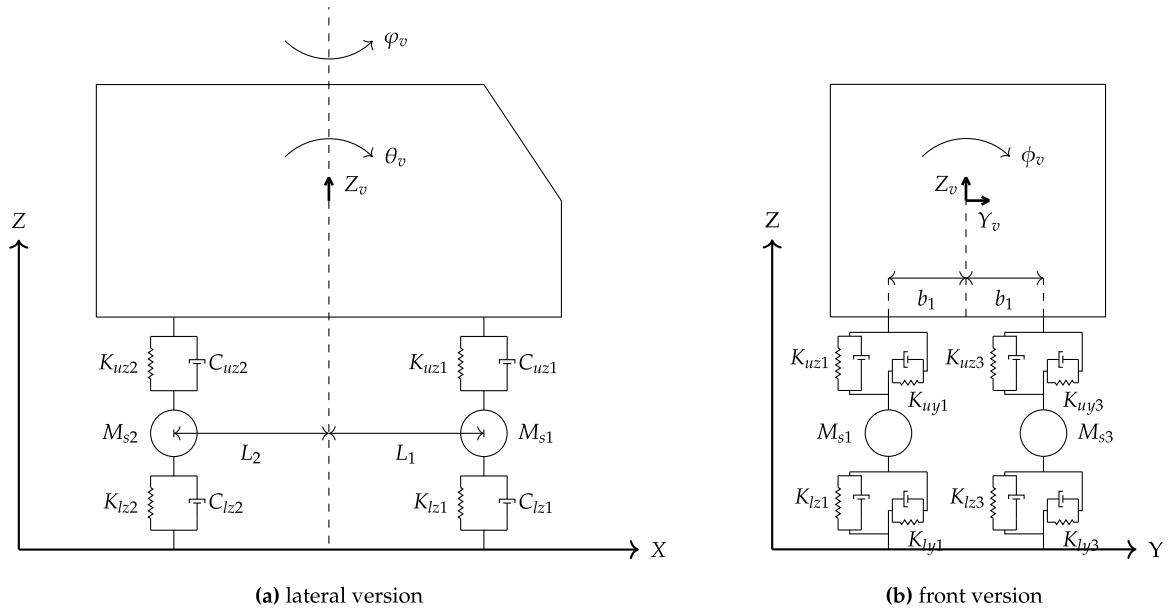


Fig. 1. 13-DOFs car model.

Table 1
Vehicle model parameters.

Model parameter	Parameter explanation	Unit	Value
M_v	Mass of rigid body	kg	1600
J_y	Pitching moment of inertia of rigid body	kg m ²	1850
J_x	Rolling moment of inertia of rigid body	kg m ²	506
J_z	Yawing moment of inertia of rigid body	kg m ²	25 000
$M_{si}, i = 1, 2, 3, 4$	Mass of axle	kg	39.5
$K_{uzi}, i = 1, 2, 3, 4$	Upper vertical spring stiffness	kN/m	109
$K_{uyi}, i = 1, 2, 3, 4$	Upper lateral spring stiffness	kN/m	79.5
$C_{uzi}, i = 1, 2, 3, 4$	Upper vertical damping coefficient	kN s/m	0.8
$C_{uyi}, i = 1, 2, 3, 4$	Upper lateral damping coefficient	kN s/m	0.8
$K_{lzi}, i = 1, 2, 3, 4$	Lower vertical spring stiffness	kN/m	176
$K_{lyi}, i = 1, 2, 3, 4$	Lower lateral spring stiffness	kN/m	58.7
$C_{lzi}, i = 1, 2, 3, 4$	Lower vertical damping coefficient	kN s/m	0.8
$C_{lyi}, i = 1, 2, 3, 4$	Lower lateral damping coefficient	kN s/m	0.8
L_1	Distance between front axle and rigid body	m	1.34
L_2	Distance between rear axle and rigid body	m	1.34
L_3	Longitudinal distance between rigid body and seat	m	0.9
b_1	Lateral distance between axle and rigid body	m	0.8
b_2	Lateral distance between rigid body and seat	m	0.3
h_v	Reference height	m	1.1
h_1	Vertical distance between rigid body and seat	m	0.3
A	Frontal area	m ²	1.96

amplitude. Therefore, only the vortex-excited force is considered to capture the essential dynamics.

VIV is a self-limiting wind-induced vibration that is driven by periodic aerodynamic forces resulting from the shedding of vortex patterns in the wake. This shedding occurs when the periodicity of the aerodynamic forces is close to the natural frequency of a specific mode of the bridge. The Scanlan nonlinear model (Ehsan and Scanlan, 1990) is used to describe its self-limiting characteristics.

$$F_{wb} = \rho U^2 D [Y_1 (1 - \epsilon \frac{y^2}{D^2}) \frac{\dot{y}}{U}] \quad (3)$$

In Eq. (3), ρ denotes the air density, U denotes the average wind speed, D denotes the deck depth, y denotes the vertical displacement of the VIVs and $\dot{y}$ denotes the vertical speed of the VIVs. The coefficients Y_1 and ϵ are functions of the reduced frequency K_v . They can be adjusted to generate vibrations with specific amplitudes. The bridge mode excitation results are shown in Table 6. In the calculation, realistic parameter values for vortex-excited forces are not used, as a specific wind speed corresponds to a single VIV amplitude for a given damping ratio. To generate a wide range of VIV amplitudes required for

Table 2
 ϵ value for different VIV amplitudes and mode orders.

VIV amplitude (m)	ϵ value		
	3rd VIV	4th VIV	5th VIV
0.05	27 000	26 000	28 000
0.10	6800	6650	6950
0.15	3050	3000	3150
0.20	1710	1680	1760
0.25	1100	1080	1130
0.30	762	746	782
0.35	560	550	576
0.40	430	420	442
0.45	340	332	350
0.50	276	270	284

this study, parameter values are selected from an extended range. For the simulation, Y_1 is set to 15, and ϵ is adjusted accordingly. Table 2 provides the corresponding ϵ values for various VIV amplitudes and mode orders.

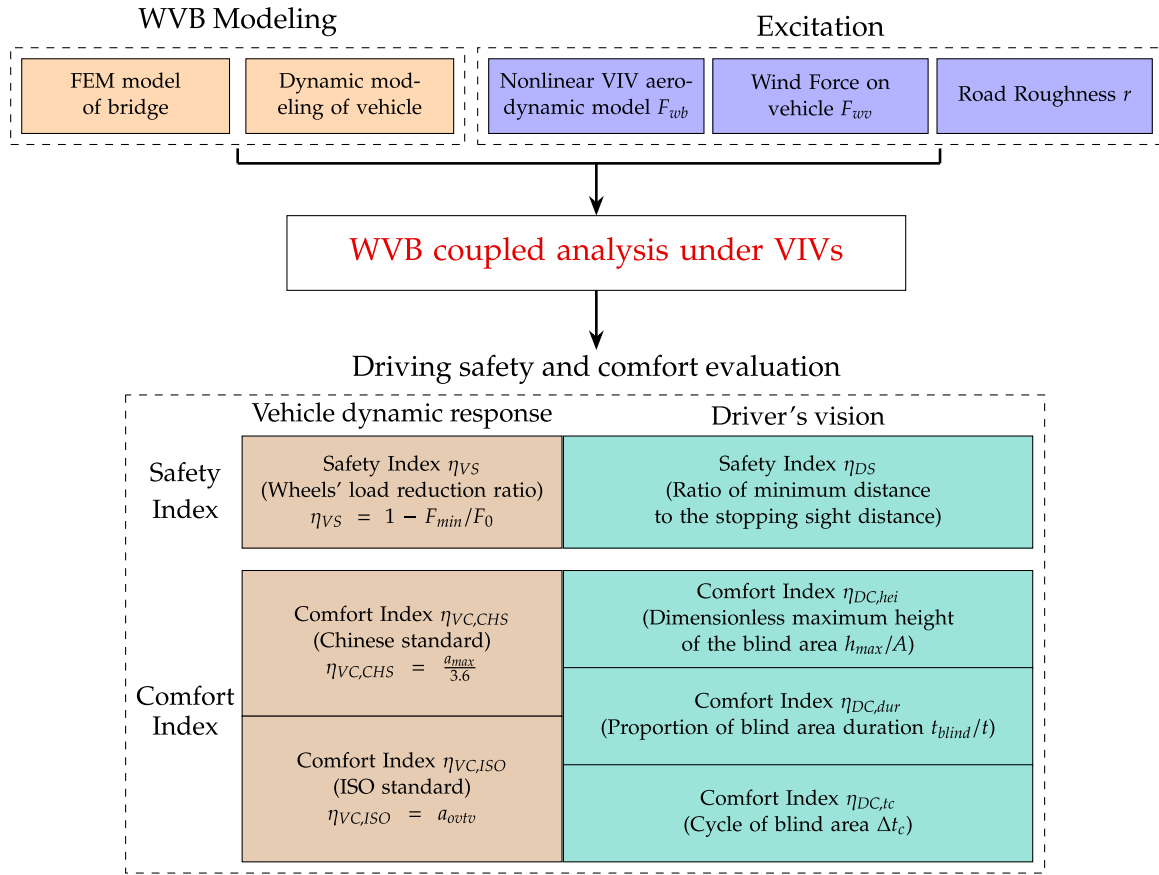


Fig. 2. Framework of driving safety and comfort evaluation by the WVBI under VIVs.

2.3. Road roughness

Road surface roughness can be expressed as a random process simulation of the power spectrum function. According to the ISO 8608 standard (Mechanical, 1995; Múčka, 2017), the road roughness power spectrum S is defined as

$$S = S_0 \cdot \frac{n^{-2}}{n_0} \quad (4)$$

In Eq. (4), S_0 is a content ($m^3/cycles$). Different values of S_0 correspond to different levels of road roughness. Because long-span suspension bridges usually fall under the management of highway departments, they generally have less roughness. In this case, only Class A is selected, with $S_0 = 4 \times 10^{-6}$. n represents the spatial frequency, with $n_0 = 0.1$ cycles/m.

After obtaining the power spectral density function, the roughness is generated using the method of inverse Fourier transformation (Huang and Wang, 1992; Camara et al., 2017).

$$r_x = \sum_{k=1}^N \sqrt{2S(n_k)\Delta n} \cos(2\pi n_k x + \phi_k) \quad (5)$$

where n_k represents the spatial frequency (m^{-1}), which is an equally spaced distribution between 0.01 and 10. Δn is the frequency resolution and ϕ_k represents a random phase ranging from 0 to 2π .

2.4. Driver's vision

When a vehicle travels through bridges during VIVs, the driver's vision may be obstructed by the peaks of the vibration wave ahead, resulting in a blind area in their vision field, as shown in Fig. 3. In this study, numerical calculations are used to obtain the bridge vibration

and vehicle vertical displacement response at the same moment, followed by the calculation of the tangent line between the driver's eye position and the bridge's vertical displacement curve. The blind area lies between the tangent line and the vertical displacement curve of the bridge. Considering the nonsmoothness of the vertical curve due to the separation between the main girder and the approach bridge, a partial approach bridge is considered in the calculation, which is assumed to not vibrate with the main bridge.

The height of the driver's eye at time t_i is

$$y_{eye}(t_i) = u_v(t_i) + h_0 \quad (6)$$

where $u_v(t_i)$ is the vertical displacement of the vehicle caused by external factors such as road roughness and bridge vibration, as calculated by Eqs. (2a) and (2b). h_0 is the vision height.

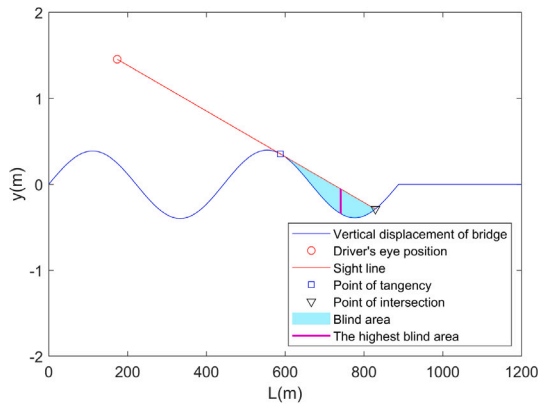
If there is a point on bridge $(x_\tau, y_b(x_\tau, t_i))$, which follows Eq. (7), this point defines the tangent line between the driver's eye position and the vertical curve of the bridge

$$\frac{y_b(x_\tau, t_i) - y_{eye}(t_i)}{x_\tau - x_v} = \left. \frac{\partial y_b(x, t_i)}{\partial x} \right|_{x=x_v} \quad (7)$$

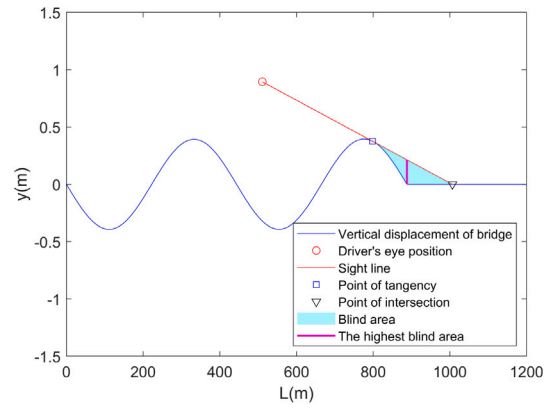
The tangent line is extended to intersect with the vertical curve of the bridge at point $(x_I, y_b(x_I, t_i))$, with coordinates following Eq. (8).

$$\frac{y_b(x_I, t_i) - y_{eye}(t_i)}{x_I - x_v} = \frac{y_b(x_\tau, t_i) - y_{eye}(t_i)}{x_\tau - x_v} \quad (8)$$

If both the tangent point $(x_\tau, y_b(x_\tau, t_i))$ and the intersection point $(x_I, y_b(x_I, t_i))$ exist, $x_\tau < x_I$ indicates the presence of a blind area lying between x_τ and x_I , as shown in Fig. 3. During vehicle movement, multiple blind areas may coexist, and blind areas closer to the vehicle have a greater influence on the driver. Therefore, only the nearest blind area is considered.



(a) Blind Area Generated in the Main Girder



(b) Blind Area Generated in the Approach Bridge

Fig. 3. Layout of blind areas while driving under VIVs.

3. Evaluation indices under VIVs

3.1. Evaluation based on the vehicle dynamic response

This study utilizes the load reduction ratio of the wheel as a safety index based on the vehicle's dynamic response, particularly in relation to overturning accidents, as described by Baker (1987). This safety index is represented by Eq. (9).

$$\eta_{VS} = 1 - \frac{F_{min}}{F_0} \quad (9)$$

where F_{min} is the minimum wheel load when passing through bridges, and F_0 is the wheel load when the vehicle is in a stationary state. The driving safety decreases as the load reduction ratio increases. When the load reduction ratio η_{vs} is at least 1, the wheel loses contact with the road surface, representing a dangerous situation for the vehicle. In the vehicle model used in this study, the rear wheel load reduction ratio always exceeds the front wheel load reduction ratio. Therefore, the safety index is represented by the rear wheel load reduction ratio in this study.

When assessing driving comfort based on vertical acceleration, a larger root mean square or peak value of vertical acceleration indicates poorer driving comfort. The Wind-resistant Design Specification for Highway Bridges (JTG/T 3360-01-2018) (Ministry of Transports of the People's Republic of China, 2018) specifies that the peak value of vertical acceleration should not exceed 3.6 m/s^2 . In the following parts, the acceleration peak is normalized by 3.6 m/s^2 as the comfort index according to the Chinese standards, known as $\eta_{VC,CHS}$. A value greater than 1 indicates that the driving comfort does not meet the Chinese standard.

$$\eta_{VC,CHS} = \frac{a_{max}}{3.6} \quad (10)$$

The ISO 2631-1:1997 standard (Mechanical Vibration, 1997) (called the 'ISO standard' hereafter) is applicable to motions transmitted to the human body through different supporting surfaces. For vehicle occupants in a sitting posture, the three supporting surfaces are the seat, floor and seat-back, as shown in Fig. 4. Each contact surface experiences vibrations in three DOFs (longitudinal, lateral and vertical) as well as three rotational DOFs). Therefore, the evaluation of vehicle occupants should involve a total of 12 DOFs. The longitudinal DOFs are mainly affected by the acceleration and deceleration of vehicles, which is not considered in the driving comfort assessment in the present study. In addition, the yawing effect of the vehicle is typically much less significant than the rolling and pitching effects (Zhou and Chen, 2016; Zhu et al., 2018). Therefore, a total of 8 DOFs are considered in the driving comfort evaluation, including a_{zs} , a_{zb} , a_{zf} , a_{ys} , a_{yb} , a_{yf} , a_p and

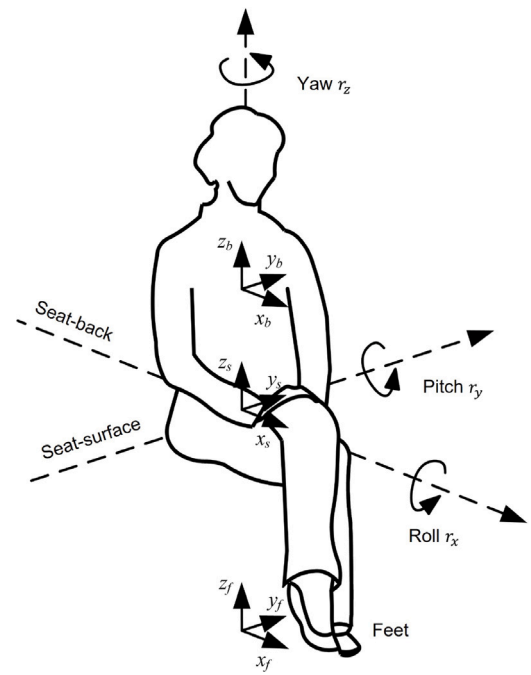


Fig. 4. Basic centric axes of the human body in the seated position.

a_r . The response direction can be vertical (v), lateral (l), pitching (p) or rolling (r). The axis locations can be the seat (s), backrest (b) or floor (f). The acceleration response at each participating axis of the vehicle can be expressed in Eqs. (11a)–(11d).

$$a_{zs} = a_{zb} = a_{zf} = \ddot{Z}_v \quad (11a)$$

$$a_{ys} = a_{yb} = a_{yf} = \ddot{Y}_v \quad (11b)$$

$$a_p = \ddot{\theta}_v \cdot L_3 \quad (11c)$$

$$a_r = \ddot{\phi}_v \cdot b_2 + \frac{1}{2} \ddot{\phi}_v \cdot h_1 \quad (11d)$$

According to the standard, the acceleration response of each DOF should first be frequency weighted to consider the influence of different vibration frequencies on the human body. After applying the frequency weighting factors, the root-mean-square (RMS) values of each DOF can be obtained. The overall vibration total value (OVTV) is the sum of the weighted RMS values multiplied by the position weighting coefficients. The criteria of different discomfort levels divided by the OVTV are

Table 3

Subjective criteria for ride comfort based on OVTV.

Overall vibration total value (m/s^2)	Subjective indication
Less than 0.315	Not uncomfortable
0.315 to 0.63	A little uncomfortable
0.5 to 1	Fairly uncomfortable
0.8 to 1.6	Uncomfortable
1.25 to 2.5	Very uncomfortable
Larger than 2	Extremely uncomfortable

Table 4

Evaluation indices based on vehicle dynamic response.

	Content of Evaluation	Limit
Safety index	Load reduction ratio of wheel $\eta_{VS} = 1 - \frac{F_{min}}{F_0}$	<1.0
Comfort indices	Comfort index based on Chinese standard $\eta_{VC,CHS} = \frac{a_{max}}{3.6}$	<1.0
	Comfort index based on ISO standard $\eta_{VC,ISO} = a_{outv}$	<1.0

shown in Table 3. In the present study, the comfort index based on the ISO standard was evaluated using the threshold of the ‘fairly uncomfortable’ stage (i.e., $a_{outv} < 1.0 \text{ m/s}^2$).

The driving safety and comfort indices based on the vehicle dynamic response are summarized in Table 4.

3.2. Evaluation based on driver's vision

The impact of blind areas can be categorized into two main effects. Firstly, they obstruct the driver's view, preventing a clear understanding of the road conditions ahead which may affect driving safety. Secondly, they contribute to discomfort or anxiety, thereby negatively affecting driving comfort. This study incorporates the concept of stopping sight distance, which assumes that if the driver can maintain clear vision within this distance, the driving scenario can be deemed safe. Any factors beyond this scope are treated as elements of driving comfort rather than safety.

To assess the influence of blind areas on driving safety, the vibrating bridge is regarded as a dynamic vertical curve of the road. The safety index η_{DS} is defined based on the stopping sight distance, as shown in Eq. (12). The distance $x_{0.1}(t)$, which represents the distance between the driver and blind area at a height of 0.1 m, must always exceed the stopping sight distance $x_{stop}(v)$.

$$\eta_{DS} = \frac{x_{stop}(v)}{\min(x_{0.1}(t))} < 1 \quad (12)$$

According to the Technical Standard of Highway Engineering (JTG B01-2014) (Ministry of Transports of the People's Republic of China, 2014), the vision height used to calculate the stopping sight distance for vehicles is 1.2 m. In the subsequent calculations of the driver's vision, the vision height is assumed to be 1.2 m by default (i.e., $h_0 = 1.2 \text{ m}$ in Eq. (6)).

For driving comfort, this study proposed three indices to comprehensively describe the influence of driver vision: the dimensionless maximum height of blind areas, the proportion of blind area duration and the change cycle of blind areas. The dimensionless maximum height of the blind area is the ratio of the maximum height of the blind area to the VIV amplitude (h_{max}/A). The proportion of blind area duration (t_{blind}/t) refers to the ratio of the time when a blind area exists to the total travel time. The change cycle of blind areas (Δt_c) represents the duration between extreme points of the maximum height of the blind area during vehicle travel. The driving safety and comfort indices based on the driver's vision are summarized in Table 5.

Table 5

Evaluation indices based on driver's vision.

	Content of evaluation	Limit
Safety index	Ratio of minimum distance between driver and blind area to stopping sight distance η_{DS}	<1.0
Comfort indices	Dimensionless maximum height of blind area $\eta_{DC,hei} = h_{max}/A$	–
	Proportion of blind area duration $\eta_{DC,dur} = t_{blind}/t$	–
	Cycle of blind area $\eta_{DC,c} = \Delta t_c$	–

Table 6

Mode diagram of the suspension bridge.

VIV frequency	Mode configuration
3rd-order vertical (0.23 Hz)	
4th-order vertical (0.27 Hz)	
5th-order vertical (0.37 Hz)	

4. Evaluation of driving safety and comfort based on vehicle dynamic response

4.1. Engineering background

Using a large-span suspension bridge as the case study, the elevation view of the bridge is shown in Fig. 5, and its mode shapes are shown in Table 6.

On May 5, 2020, the bridge experienced a significant VIV from 12:30 to 18:30 (Ge et al., 2022; Zhao et al., 2022). According to the monitoring data, the VIVs mainly manifested as vertical vibrations of the bridge. During the deployment of the temporary water-filled barriers, the vertical vibration frequency was 0.37 Hz, corresponding to the 5th order vertical mode, with a maximum amplitude of approximately 31 cm. The wind speed during the vibration was approximately 10–12 m/s. After removing the temporary water-filled barriers, the large-amplitude vibration was suppressed. However, the secondary VIVs exhibited 4th order and 3rd order vertical modes, with amplitudes ranging from 3–20 cm. The wind speed during the vibrations was approximately 7–9 m/s.

4.2. The influence of VIV modes

Driving safety and comfort decrease as the order of the VIV mode increases. The illustration in Fig. 6 demonstrates the influence of various modes of VIVs on the vehicle dynamic response at an amplitude of 0.3 m and a vehicle speed of 60 km/h. The improvement in relevant indices for the 5th order VIV compared to the 3rd order VIV is at least 60%. Under the same conditions, the ISO comfort index has stricter requirements. The ISO comfort index remains within limits when 4th order VIV occurs but exceeds limits by 35% when the 5th-order VIV occurs. However, the Chinese standard comfort index remains within limit under the 5th-order VIV.

4.3. The influence of vehicle speed and VIV amplitude

Taking the unfavorable 5th order VIV as an example, Figs. 7, 8 and 9 show that increasing vehicle speed and VIV amplitude have adverse effects on both driving safety and comfort. For the VIV amplitude, Fig. 8

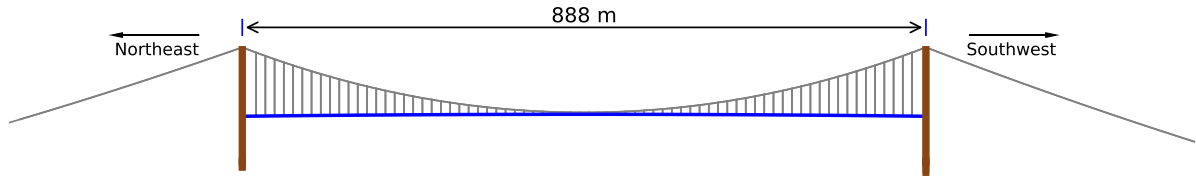


Fig. 5. Elevation view of the long-span suspension bridge (unit: m).

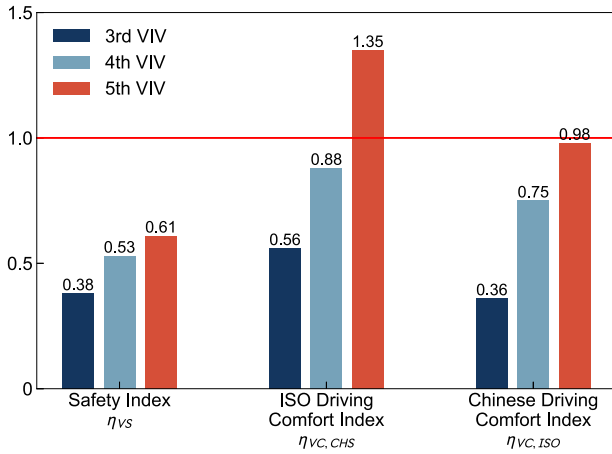


Fig. 6. The influence of VIV modes at a speed of 60 km/h and a 0.3 m vibration amplitude.

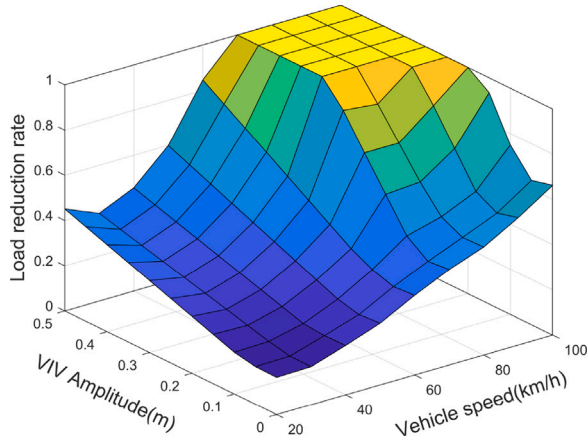


Fig. 7. The influence of vehicle speed and VIV amplitude on driving safety under 5th-order VIV.

presents the calculated results of vertical displacement and vertical acceleration of the vehicle at different amplitudes. The results indicate that an increase in VIV amplitude significantly increases the vertical acceleration of the vehicle, leading to a substantial decrease in driving comfort. With an amplitude of 0.3 m, the comfort index at a vehicle speed of 60 km/h precisely meets the requirements of the Chinese standard. However, at a vehicle speed of 80 km/h, the Chinese standard comfort index is 1.9, exceeding the limits of 90%.

As the vehicle speed increases, the sensitivity of the vehicle dynamic response to bridge vibration rapidly increases, indicating significant nonlinear characteristics, as illustrated in Fig. 10. For lower vehicle speeds, the ISO comfort index experiences relatively gradual growth as the VIV amplitude increases. However, at higher vehicle speeds, the ISO comfort index changes rapidly with the vehicle speed, exceeding 2 m/s^2 at an amplitude of approximately 0.2 m, reaching the range classified as 'extremely uncomfortable'. When the actual vibration amplitude is

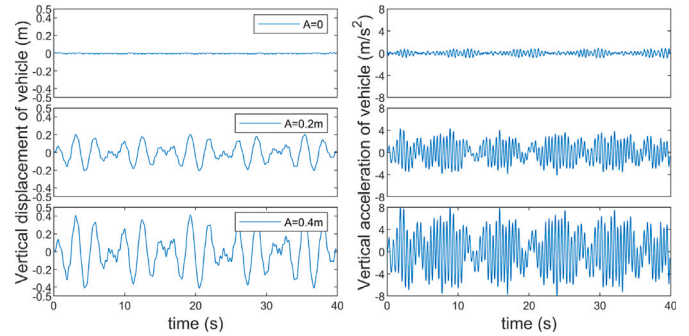


Fig. 8. Time-history curve for dynamic response of vehicle.

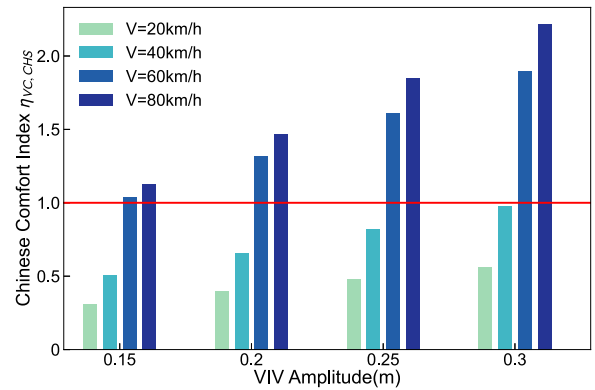


Fig. 9. The influence of vehicle speed and VIV amplitude on the Chinese driving comfort index under 5th order VIVs.

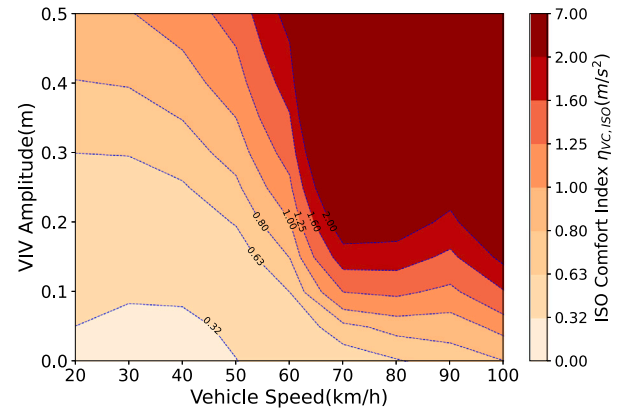


Fig. 10. The influence of vehicle speed and VIV amplitude on the ISO driving comfort index under 5th order VIV.

0.31 m, maintaining the vehicle speed below 50 km/h ensures that the driving comfort index remains within the 'fairly uncomfortable' range.

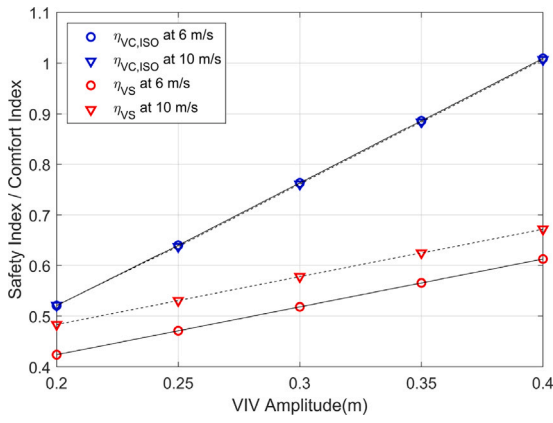


Fig. 11. Indices at 6 m/s and 10 m/s under the 4th order VIV (driving speed 60 km/h).

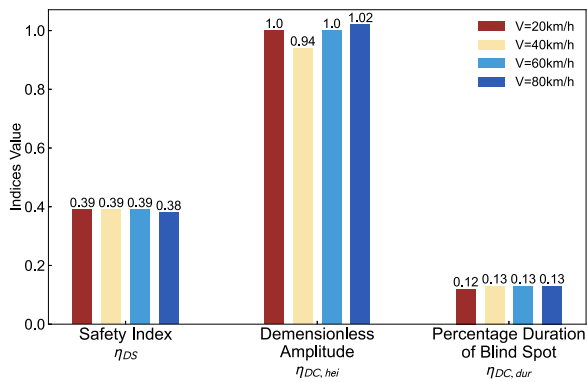


Fig. 12. The influence of vehicle speed on indices based on driver vision.

4.4. The influence of the crosswind effect

During VIVs, the turbulence intensity of the crosswind is low. Therefore its static impact on vehicles is greater than its dynamic effect, which primarily affects the driving safety of the vehicle. Fig. 11 compares the driving safety index and the ISO comfort index under 6 m/s and 10 m/s wind speed with turbulence of 0.1. The influence of the crosswind on the ISO comfort index is negligible, but it can cause an increase in the wheel load reduction rate of the wheels on the windward side, resulting in the deterioration of driving safety.

5. Evaluation of driving safety and comfort based on driver vision

5.1. The influence of vehicle speed

The variations in the driving safety and comfort indices based on the driver's vision were calculated at different vehicle speeds, as shown in Fig. 12. The relative error between the results of the same index at different speeds did not exceed $\pm 5\%$, indicating that the variation in speed had little influence on the indices based on the driver's vision. The slight deviations at different speeds can be attributed to the different vertical dynamic responses of the vehicle caused by different driving speeds, as shown in Eq. (6). An increase in vehicle speed leads to a corresponding increase in the vertical acceleration of the vehicle. However, its impact on the vertical displacement is relatively small and thus does not affect the driver's vision.

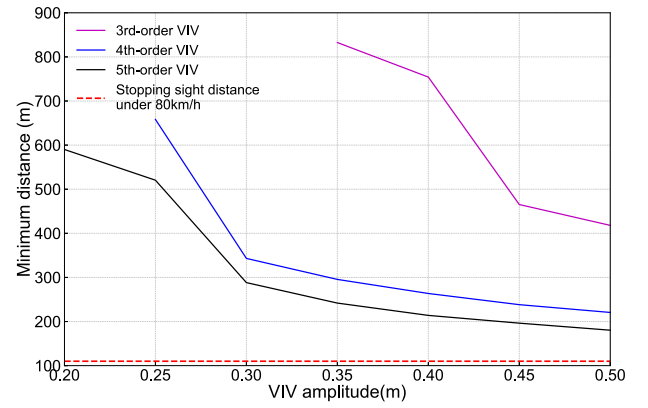


Fig. 13. Minimum distance between the driver and blind area at a height of 0.1 m.

5.2. Safety index evaluation

When the vehicle speed is 80 km/h and the VIV amplitude reaches 0.5 m, the minimum distance between the driver and the blind area at a height of 0.1 m, hereafter referred to as the 'minimum distance', still exceeds the stopping sight distance, as shown in Fig. 13. Therefore, the safety index based on the driver's vision consistently meets the requirements. In Fig. 13, the minimum distance of the 4th and 5th order VIVs experiences a sudden change at an amplitude of 0.3 m. When the VIV amplitude is less than 0.3 m, the maximum height of the blind area at the adjacent vibration half-wave does not exceed 0.1 m. Instead, it reaches 0.1 m at a further vibration half-wave. Fig. 14 illustrates the variation in the blind area position when the VIV amplitude is 0.25 m and 0.3 m at the same moment. When the amplitude is 0.25 m, almost no blind area is produced at the adjacent vibration half-wave, and a blind area whose maximum height exceeds 0.1 m occurs at a further vibration half-wave. When the amplitude is 0.3 m, the maximum height of the closest blind area exceeds 0.1 m, resulting in a significant decrease in the minimum distance. This conclusion is consistent with the findings of Chen and Huang (2015), who used a simplified method that calculates only the maximum height of the closest blind area.

When the VIV amplitude exceeds 0.3 m, a clear functional relationship between the minimum distance and the VIV amplitude emerges. Polynomial fitting is performed using the VIV amplitude A and the approximate half-wavelength of the bridge d as independent variables.

$$x_{min} = \sum_{j=0}^{m_1} \sum_{i=0}^{m_2} a_i d^i A^j \quad (13)$$

In Eq. (13), m_1 represents the polynomial order of the VIV amplitude A , while m_2 represents the polynomial order of the approximate half-wavelength of the bridge d .

When $m_1 = 2$ and $m_2 = 1$, the fitting equation can be proposed as

$$x_{min} = (9.32d + 21.9)A^2 + (-10.81d + 100.7)A + (4.07d - 54.3) \quad A \in [0.3, 0.5] \quad (14)$$

In Eq. (14), x_{min} is the minimum distance. A represents the VIV amplitude. d represents the approximate half-wavelength of the bridge, which can be calculated by Eq. (15).

$$d = \frac{L}{n} \quad (15)$$

where L is the span of the bridge, and n is the mode order during VIVs.

Given that the blind area are primarily affected by the vertical vibration displacement of the vehicle, the key factors are bridge span, VIV amplitude and VIV mode. Other factors, such as road roughness, have a negligible impact on the vertical vibration displacement. Therefore, Eq. (14) is widely applicable to other bridges. The fitting error of

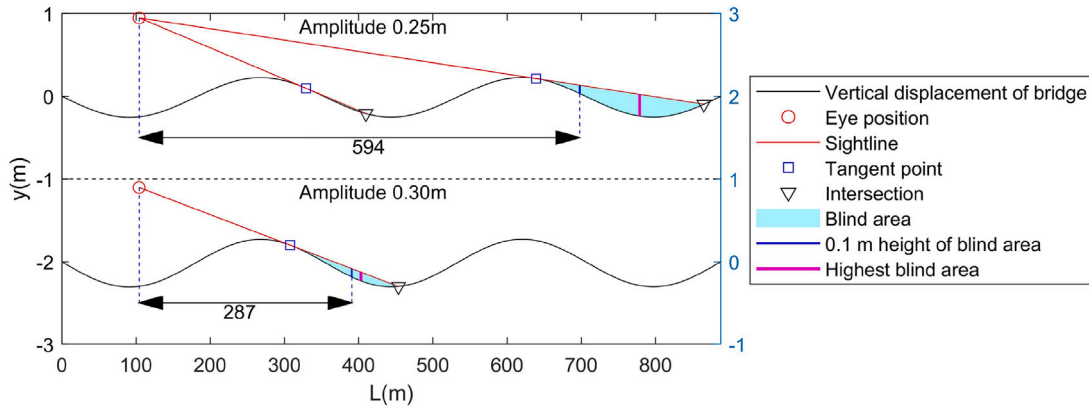


Fig. 14. Comparison of the blind areas at certain moments with VIV amplitudes of 0.25 m and 0.3 m.

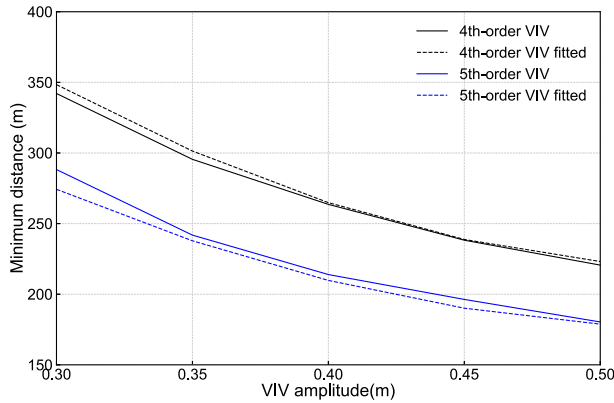


Fig. 15. Fitting results for the minimum distance.

Eq. (14) is within the range of $\pm 6\%$, as shown in Fig. 15. It mainly arises from the discrepancy between the actual vibration half-wavelength and the approximate half-wavelength calculated by Eq. (15). Because each wave is assumed to have the same amplitude under one VIV mode, this fitting equation is not applicable to a small number of VIV modes, similar to the 3rd order VIV in this case.

5.3. Comfort index evaluation

The cycle of the blind area $\eta_{DC,tc}$ is influenced mainly by the VIV mode and is independent of the vehicle speed and VIV amplitude. Fig. 16 illustrates the time history of the maximum height of the blind area as vehicles pass over the bridge under the 4th order VIV with a speed of 40 km/h and a VIV amplitude of 0.4 m. This study divides the time history of the blind area into four stages. In Stage 1, a significant blind area can be observed in the main girder, irrespective of the position of the vehicle at the wave crest or trough of the bridge vibration. In Stage 2, at the trough of the vibration, the blind area occurs in the main girder. At the crest of the bridge vibration, the blind area occurs in the approach bridge. In Stage 3, the blind area, which is on the approach bridge, no longer occurs at the crest of the bridge vibration but only when the vehicle is at the trough of the bridge vibration. In Stage 4, no blind area occurs as the vehicle approaches the connection between the main bridge and the approach bridge. In this study, the end of Stage 1 is defined as the time of the last extremum point before the blind area on the approach bridge is generated. The end of Stage 2 is defined as the time of the first extremum point after the last blind area on the main girder is generated. Finally, the end of Stage 3 is defined as the time of the last extremum point of the blind area. In Stages 1 and 2, the cycle of the blind area $\eta_{DC,tc}$ is half of the

VIV period. In Stage 3, the change cycle is equal to the period of VIV. This result is independent of the vehicle speed and VIV amplitude.

When the blind area is fully developed (the VIV amplitude exceeds 0.25 m), the approximate positions of each stage can be determined based on the vertical displacements of the vehicles in each stage. Assuming that n th-order VIV ($n \geq 4$) occurs, i.e., the bridge has n vibration half-waves, the vehicles traveling from the first to $n - 3$ th vibration half-waves correspond to Stage 1, the $n - 2$ th vibration half-wave corresponds to Stage 2, the $n - 1$ th vibration half-wave corresponds to Stage 3, and the last vibration half-wave corresponds to Stage 4, as shown in Fig. 17. If 3rd order VIV occurs, only Stages 2, 3 and 4 will occur. If 2nd order VIV occurs, only Stages 3 and 4 will occur. If 1st order VIV occurs, no blind area will emerge.

The amplitude of the VIV affects the duration of each stage, as shown in Fig. 18. When the VIV amplitude decreases, the durations of the first three stages are shortened, resulting in improved driving comfort. When the VIV amplitude exceeds 0.25 m, changes in amplitude have a relatively minor influence on the proportion of stage durations. Conversely, when the VIV amplitude is less than 0.25 m, the first three stages may deteriorate or vanish due to the excessively low VIV amplitude.

The duration of the blind area was positively correlated with both the amplitude and VIV mode. A larger VIV amplitude and higher mode order of VIV would result in a longer duration of the blind area, as illustrated in Fig. 19. When comparing VIVs of the same amplitude, the duration of the blind area for the 5th order VIV is longer than that for the 4th order VIV, and this difference becomes more pronounced with increasing VIV amplitude. The maximum height of the blind area increases with the amplitude, as shown in Fig. 19. The influence of the VIV mode on the maximum height of the blind area is relatively weak. When the amplitude is less than 0.4 m, the maximum height of the blind area under the 5th order VIV is slightly greater than that under the 4th order VIV. However, when the amplitude exceeds 0.4 m, the opposite occurs, and the differences between the two modes are small. Therefore, no direct relationship exists between the VIV mode and the maximum height of the blind area.

5.4. The influence of vision height

According to the Technical Standard of Highway Engineering (JTG B01-2004) (Ministry of Transports of the People's Republic of China, 2014), the standard visual height for passenger cars is 1.2 m, while for heavy trucks it is 2.0 m. To ensure our calculations comprehensively cover a variety of vehicle types and scenarios, we selected 1.6 m as an intermediate value between passenger cars and heavy trucks.

A larger visual height is beneficial for both driving safety and comfort. Fig. 20 illustrates the influence of the vision height under the 5th order VIV, with a vehicle speed of 80 km/h and a VIV amplitude of

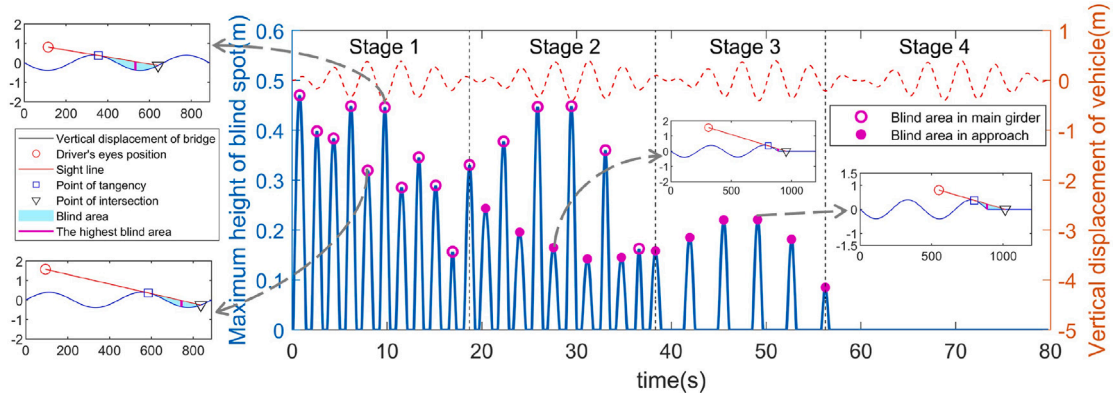


Fig. 16. Time history of the maximum height of the blind area at a vehicle speed of 40 km/h and a VIV amplitude of 0.4 m under the 4th order VIV.

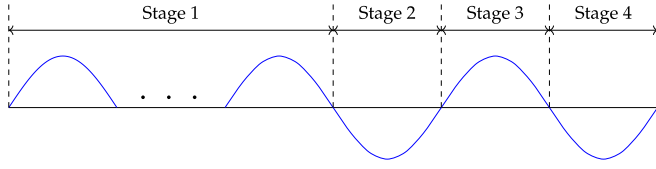


Fig. 17. Stage definition when blind area fully develops.

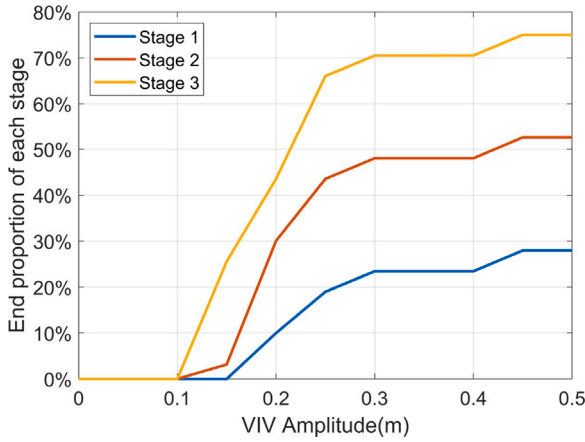


Fig. 18. The influence of the VIV amplitude on the end proportion of each stage under the 4th-order VIV at a speed of 40 km/h.

0.3 m. As the height increases, the position of the blind area at a height of 0.1 m moves away from the driver, resulting in an improvement in the safety index. Additionally, the dimensionless maximum height of the blind area and the duration of the blind area decrease as the vision height increases, resulting in an improvement in driving comfort. A greater vision height does not change the cycle of the blind area, but it can influence the proportion of individuals in each stage. Under the same amplitude, the proportion of the high-frequency range (Stage 1 plus Stage 2) decreases significantly as the vision height increases, which implies that the driver could enter Stage 3 earlier. In this stage, $\eta_{DC,tc}$ is larger, which enhances driving comfort.

6. Discussion of VIV amplitude limits

The results of the VIV amplitude limits are compared by considering the vehicle dynamic response, the driver's vision and the method provided by the wind-resistance design specification (Ministry of Transport of the People's Republic of China, 2018). The comfort indices based on the driver's vision currently do not have clear limits and are

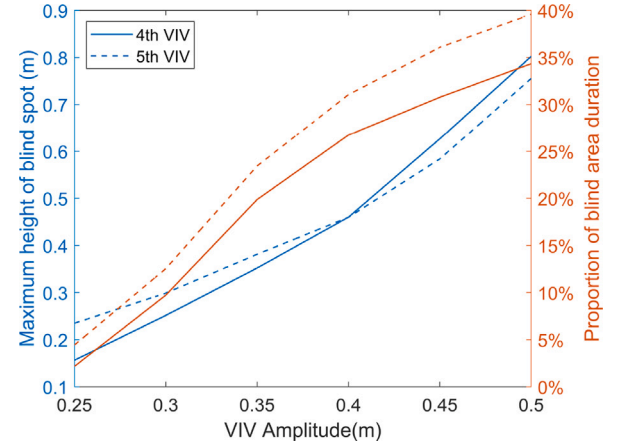


Fig. 19. The influence of VIV amplitude and VIV mode on driving comfort indices based on drivers' vision.

therefore not considered in the amplitude limit evaluation. As shown in Fig. 21, compared to the safety index based on the dynamic response and the driver's vision, the driving comfort index is the controlling index determining the permitted VIV amplitude. The VIV amplitude limit based on the driving comfort index is related to the driving speed. The speed at which the amplitude limit obtained from the ISO comfort index is equal to that calculated by specification is defined as the 'critical speed'. For this 5th order VIV, the critical speed is approximately 68 km/h. When the vehicle speed is lower than the critical speed, the amplitude limit obtained by the specification is lower and therefore provides guidance for controlling the VIV amplitude limit. However, when the vehicle speed exceeds the critical speed, the amplitude limit obtained via specification becomes ineffective. According to the conclusion in Section 4.2, an increase in the mode order of the VIV or a decrease in bridge span is detrimental to driving safety and comfort, resulting in a decrease in the critical speed and the speed range controlled by the specifications.

The VIV amplitude limits are highly relevant to the vehicle speed. This study suggests VIV amplitude limits at different speeds based on driving comfort, as illustrated in Fig. 22. The fitting equations are provided following the specification format as follows:

$$A_{limit} = \frac{0.095}{f_b} \quad \text{when } v = 60 \text{ km/h} \quad (16a)$$

$$A_{limit} = \frac{0.028}{f_b} \quad \text{when } v = 80 \text{ km/h} \quad (16b)$$

$$A_{limit} = \frac{0.025}{f_b} \quad \text{when } v = 100 \text{ km/h} \quad (16c)$$

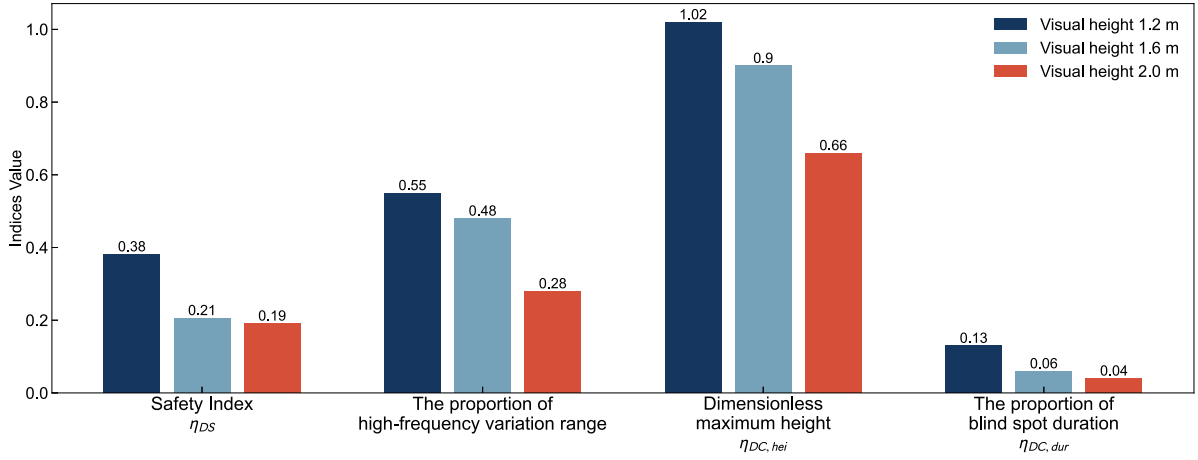


Fig. 20. The influence of vision height on driver vision under a 5th order VIV at an amplitude of 0.3 m and a speed of 80 km/h.

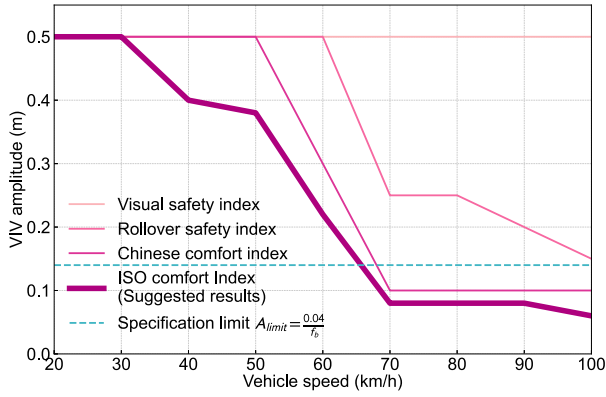


Fig. 21. Comparison of VIV amplitude limits for different indices (using 5th order VIV as an example).

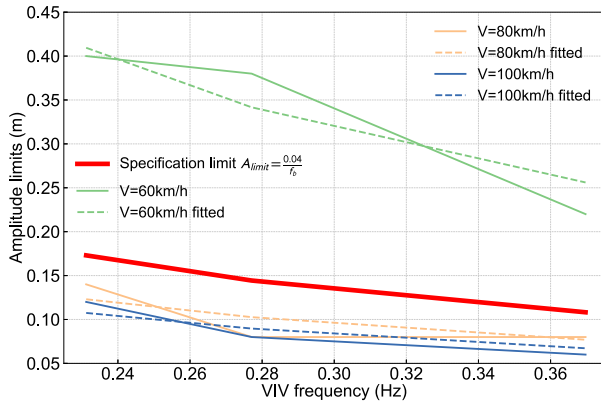


Fig. 22. VIV amplitude limits under different vehicle speeds and fitting results.

Combining the above equations and using the vehicle speed as a coefficient, the relationship between the numerator from Eqs. (16a) to (16c) and the vehicle speed is shown in Fig. 23. The fitting result is

$$A_{limit} = \frac{av^2 + bv + c}{f_b} \quad (17)$$

where v represents the driving speed, which ranges from 40 km/h to 100 km/h. The coefficients in this speed range are $a = 5.721 \times 10^{-5}$, $b = -0.011$, and $c = 0.596$. f_b is the frequency of VIV. When the driving speed is less than 40 km/h, the VIV amplitude limits based on driving comfort are so large that structural safety might be the controlling

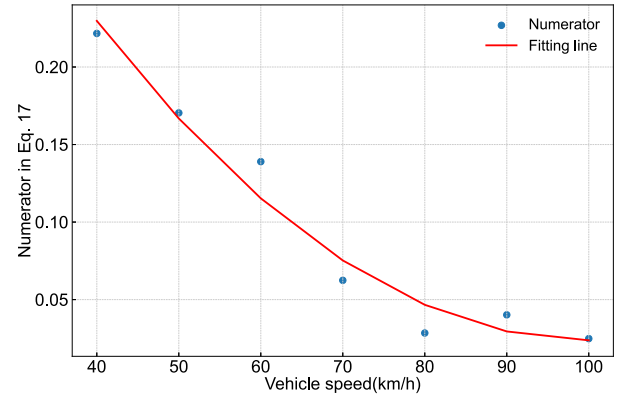


Fig. 23. Fitting result between the numerator and vehicle speed.

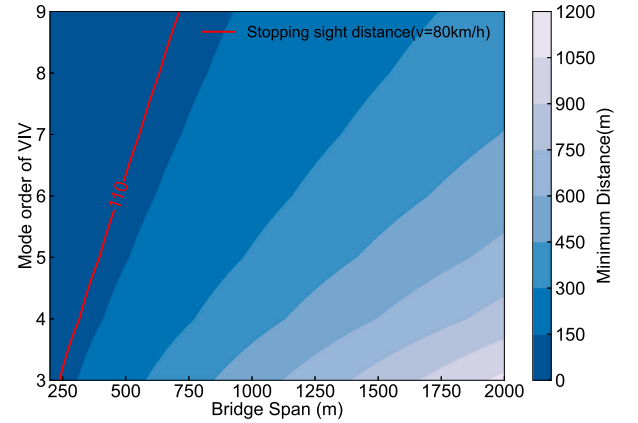


Fig. 24. Minimum distance variation under different spans and VIV modes with an amplitude of 0.3 m.

factor.

For visual safety, Fig. 24 illustrates the distribution of the minimum distance with respect to the bridge span and mode order of the VIV. For a VIV amplitude of 0.3 m, $\eta_{DS} > 1$ is only possible for smaller bridge spans and higher mode orders of VIV. This situation can only occur for large-span bridges under extreme high-order VIV (larger than 10th order VIVs). Therefore, vision safety is highly unlikely to become a controlling factor.

7. Conclusions

The comprehensive evaluation of driving safety and comfort under VIVs involves two perspectives: the vehicle's dynamic response and the driver's vision. For the first time since the discovery of the VIVs induced blind area, its development and influence on driving safety have been quantified. Various factors influencing driving safety and comfort are considered and a methodology for incorporating driving speed into the process of setting VIV amplitude limits are proposed. Some key conclusions can be summarized as follows:

(1) Based on a comprehensive comparison of different indices, including overturning safety, driving comfort based on vehicle dynamic response and vision safety, driving comfort based on vehicle dynamic response serves as the primary index for controlling vehicle speed under VIV conditions.

(2) Increases in the vehicle speed, amplitude and mode order of VIV have detrimental effects on driving safety and comfort. A surge of over 25% in the ISO driving comfort index is anticipated with each incremental increase in VIV modes at high speeds. Under VIV conditions, mild crosswind conditions primarily affect driving safety and have little influence on driving comfort.

(3) This study incorporates the concept of stopping sight distance to quantify the influence of blind area on driving safety and introduces a fitting equation for driving safety based on drivers' vision, demonstrating broad applicability. Based on the analysis, it is almost impossible for vision safety to be the controlling index for large span bridges. The development of blind areas under VIVs can be categorized into four stages. The time interval between changes in the blind area is related only to the natural vibration period. The amplitude of the VIV affects the duration of each stage, and the driving speed has no influence on the time interval of the blind area.

(4) The VIV amplitude limits, which are determined by the comfort index based on the vehicle dynamic response, are related to the driving speed. The VIV amplitude limit calculated according to the specifications becomes ineffective when the vehicle speed exceeds the critical speed. The critical speed decreases as the VIV mode order increases or the bridge span decreases. Furthermore, this study referenced the specification format to provide VIV amplitude limits considering driving comfort based on vehicle dynamic response at different vehicle speeds.

The conclusions of this paper are currently derived from the specific case of the Humen Bridge, which is a suspension bridge with a main span of 888 m. Due to the lack of limits for driving comfort based on driver vision, this study describes only the changes in the trends of indices without considering those within the limits of VIV amplitude.

In the future, comprehensive evaluations of driving comfort under VIV conditions based on driving simulation devices, the collection of physiological signals of drivers during VIVs, and objective driving comfort indices to determine the limits of VIV amplitude are expected.

CRediT authorship contribution statement

Yiheng Fu: Writing – original draft, Software, Methodology. **Fengying Wu:** Writing – review & editing, Methodology. **Chen Chai:** Writing – review & editing. **Wei Cui:** Writing – review & editing. **Yongle Li:** Writing – review & editing. **Lin Zhao:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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